MCKV INSTITUTE OF ENGINEERING

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Curriculum for Undergraduate Degree (B.Tech.) in Computer Science and Engineering (AI and ML) (w.e.f. AY: 2022-23)

Part III: Detailed Curriculum

FIFTH SEMESTER

Course Name:	**Machine Learning**		
Course Code:	PC-CS(D) 501	**Category:**	Professional Core Courses
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	BSM 301, BSM 40404
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To learn the concepts of data and patterns
2	To design and analyze various machine learning algorithms.
3	Explore supervised and unsupervised machine learning
4.	Explore Deep Learning Techniques and various feature extraction.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Supervised Learning: - Distance based methods, Nearest Neighbor. Learning Techniques with Decision Tree and Naïve Bayes Classifier.	6
2	Supervised Learning (Regression/Classification):- Linear Regression, Logistic Regression, Linear Models optimization SVM, Dealing with Non Linearity and Kernel Methods. Multi class classification, Ranking	8
3	Introduction to Unsupervised Learning:- K-Means Clustering, Kernel K-Means, Dimensionality Reduction with PCA and Kernel PCA. Preliminary idea of Factorization and generative models (Mixture model and Latent factor model).	8
4	Evaluating Machine Learning Algorithms model selection, Introduction to statistical learning theory and Ensemble Methods (Bagging, Boosting and Random Forests).	6
5	Model Estimation, Modeling Time Series Data, Deep Learning and Feature Extraction Techniques. Shallow Neural Network and Deep Neural Network.	7
6.	Case Study: - Selection from a Technique and implementation with Model Selection.	1
Total		**36**

Course Outcomes:	
1.	Use different supervised learning strategies to infer a uniclass and multiclass model.
2.	Show the difference between Linear and Non-Linear model with different dimensional reduction techniques.
3.	Implement different unsupervised learning techniques.
4.	Interpret the concept of ensemble learning, model estimation and deep learning techniques.

Learning Resources:	
1	Machine Learning, Tom Mitchell, McGraw Hill, 1997.
2	Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2007
3	Kevin Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012
4	Hastie, Tibshirani, Friedman The Elements of Statistical Learning Springer 2007

Course Name:	**Artificial Intelligence**		
Course Code:	PC-CS(AM) 502	**Category:**	Professional Core Course
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	Discrete Mathematics, Algorithm
Full Marks:	70		
Examination Scheme:	Semester Examination: -70	Continuous Assessment: - 25	Attendance: -5

Course Objectives:	
1	This subject will help in acquiring knowledge on Basic Techniques of AI and Intelligent agents**.**
2	The subject will help to acquire knowledge on reasoning with and without uncertainty.
3	The subject will help in having elementary knowledge on expert system development in various domain.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction [2] Overview of Artificial intelligence- Problems of AI, AI technique, Tic - Tac - Toe problem. Intelligent Agents [3] Agents & environment, nature of environment, structure of agents, goal based agents, utility-based agents, learning agents. Problem Solving [3] Problems, Problem Space & search: Defining the problem as state space search, production system, problem characteristics, issues in the design of search programs.	8
2	Search techniques [6] Solving problems by searching: problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Heuristic search strategies [6] Greedy best-first search, A* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search, genetic algorithms; constraint satisfaction problems, local search for constraint satisfaction problems. Adversarial search [3] Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening.	15
3	Using predicate logic [2] Representing simple fact in logic, representing instant & ISA relationship, computable functions & predicates, resolution, natural deduction. Probabilistic reasoning [4]	6

	Representing knowledge in an uncertain domain, the semantics of Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics.	
4	Applications of AI in the following domain (a) E-Commerce (b) Expert Education (c) Finance (d) social media (e) Robotics. Preliminary knowledge of Chatbots and AI based Games. (f) Autonomous Vehicle	7
Total		**36L**

Course Outcomes:	
After completion of the course, students will be able to:	
1.	Identify the basic concepts of AI Techniques, Agents, and Production System.
2.	Compare between heuristic , non-heuristic search strategies and adversarial search strategies.
3.	Apply Predicate logic, Resolution and Probability based inference in different problems.
4.	Explain the expert system architecture in different domains.

Learning Resources:	
1.	Artificial Intelligence, Ritch & Knight, TMH
2.	Artificial Intelligence A Modern Approach, Stuart Russel Peter Norvig Pearson
3.	Introduction to Artificial Intelligence & Expert Systems, Patterson, PHI
4.	Poole, Computational Intelligence, OUP
5.	Expert Systems, Giarranto, VIKAS
6.	M.C. Trivedi, Artificial Intelligence, Khanna Publishing House, New Delhi (AICTE Recommended Textbook – 2018)

Course Name:	**Computer Networks**		
Course Code:	PC-CS(AM)503	**Category:**	Professional Core Courses
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	Communication Engineering and Operating Systems
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To develop an understanding of modern network design and performance.
2	To introduce the student to the major concepts involved in variety of networks.
3	To provide a foundation to apply in network programming

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Introduction:** Data communication Components, Data Representation and its flow, Networks, Topology, Protocols and Standards, OSI model, TCP/IP, Transmission Media, Switching, Connecting Devices.	5
2	**Physical Layer:** Brief of Data and Signals- Basics, SNR, Bandwidth-Delay, Nyquist theorem, Shannon's Capacity, Baud, Signal Impairments, Data to Signal conversion Techniques- Digital to Analog, Digital to Digital, Scrambling, Multiplexing.	6
3	**Data Link Layer and Medium Access Sub Layer:** Framing, Byte/ Character stuffing, Bit stuffing; Error Detection and Error Correction - Fundamentals, Hamming Distance, Parity, CRC, Checksum; Flow Control and Error control protocols – ARQ- Stop and Wait, Sliding Window- Go- Back- N, Selective Repeat, Piggybacking, HDLC, Multiple access- Pure ALOHA, Slotted ALOHA, CSMA/CD, CDMA/CA, Token Passing, Poll, TDMA, FDMA, CDMA, LAN: Wired LAN, Wireless LAN, VLAN.	9
4	**Network Layer:** Logical addressing and Protocols– IPV4, IPV6; ICMP, Address mapping – ARP, RARP, BOOTP and DHCP–Delivery, Forwarding and Unicast Routing protocols.	8
5	**Transport Layer:** Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), Congestion Control, Quality of Service, QoS improving techniques: Leaky Bucket and Token Bucket algorithm.	4
6	**Application Layer:** DNS, Telnet, Email, FTP, HTTP, Firewalls, Basics of Cryptography.	4
Total		**36L**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Classify the role of the components of computer networks using addressing mechanisms and layered models.
2	Explain variety of protocols and security techniques at application layer.
3	Compare different signal conversion techniques, transmission media, switching methodologies.
4	Analyze different network and transport layer protocols.
5	Evaluate different error, flow and access control protocols over variety of networks.

Learning Resources:	
1	"Data Communications and Networking (4th Ed.)" – B. A. Forouzan, TMH
2	"Computer Networks (4th Ed.)" – A. S. Tanenbaum, Pearson Education/PHI
3	"Data and Computer Communications (5th Ed.)"- W. Stallings, PHI/ Pearson Education

Course Name:	**High-Performance Computing**		
Course Code:	PC-CS (AM) 504	**Category:**	Professional Core Courses
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	Computer Organization and Architecture
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To know how modern high-performance processors are organized their strengths and weaknesses.
2	To study about the architecture of parallel systems.
3	To gain knowledge about the analytical parallel algorithms.

Course Contents:

Module No.	**Description of Topic**	**Contact Hrs.**
1	Introduction: - Review of basic computer architecture, quantitative techniques in computer design, measuring and reporting performance.	2
2	Levels and model of parallelism: - Parallel Processing concepts, instruction, transaction, task, thread, memory, function, data flow models, demand-driven computation;	2
3.	Parallel architectures: - superscalar architectures, multi-core, multi-threaded, server and cloud.	2
4.	Fundamental design issues in HPC: - Load balancing, scheduling, synchronization, and resource management; Operating systems for scalable HPC; Parallel languages and programming environments; OpenMP/Pthread/MPI/Java/Cilk; Performance analysis of parallel algorithms.	10
5.	Fundamental limitations in HPC: bandwidth, latency and latency hiding techniques.	1
6.	Scalable storage systems: RAID, SSD cache, SAS, SAN; HPC based on cluster, cloud, and grid computing: economic model, infrastructure, platform, computation as service	2
7.	Benchmarking HPC: scientific, engineering, commercial applications, and workloads;	2
8.	Accelerated HPC: architecture, programming and typical accelerated system with GPU, FPGA, Xeon Phi, Cell BE;	2
9.	Power-aware HPC Design: computing and communication, processing, memory design, interconnect design, power management;	2
10.	HPC programming assignments: Hands on experiment and programming on parallel machine and HPC cluster using Pthread / OpenMP/ MPI/ Nvidia Cuda /Cilk. Standard multiprocessor simulator or cloud simulator at an introductory level only.	9

11.	Distributed shared-memory architecture. Cluster computers. Non von Neumann architectures: data flow computers, reduction computer architectures, systolic architectures.	2
Total		**36L**

Course Outcomes:	
After completion of the course, students will be able:	
1	Describe design structures of pipelined and parallel processing systems.
2	Explain the design issues of parallel systems including limitations, benchmarking, and storage systems.
3	Implement different parallel programming concepts like OpenMP, MPI, NVDIA CUDA and Cilk.
4	Explain the concept of MPI, Cluster Computers, Non-Von Neuman Architecture.

Learning Resources:	
1	Georg Hager and Gerhard Wellein. Introduction to High Performance Computing for Scientists and Engineers (1st ed.). CRC Press, Chapman & Hall/CRC
2	Ananth Grama and George Karypis, “Introduction to parallel computing”, Addison -Wesley, 2009.
3	John Levesque and Gene Wagenbreth, “High Performance Computing: Programming and Applications”, Chapman & Hall, 2010.
4	John L. Hennessy and David Patterson, “Computer Architecture- A Quantitative Approach”, Elsevier, 2012.
5	Michael Quinn, “Parallel Programming in C with MPI and OpenMP”, Indian edition, McGraw Hill Education, 2017.

Course Name:	**Numerical Methods**		
Course Code:	BS-M 504	**Category:**	Basic Science Course
Semester:	5th	**Credit:**	2
L-T-P:	2-0-0	**Pre-Requisites:**	Some concepts from basic math – algebra, geometry, pre-calculus and statistics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To compute different numerical errors in computations.
2	To learn interpolation techniques.
3	To apply the techniques for solving integrations, ODEs.
4	Solve linear and non-linear equations.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Approximation in numerical computation:** Truncation and rounding errors, Fixed and floating-point arithmetic, Propagation of errors.	2
2	**Interpolation:** Newton's Forward Interpolation, Newton's Backward Interpolation, Lagrange's Interpolation, Newton's Divided Difference Interpolation	4
3	**Numerical integration:** General Quadrature Formula, Trapezoidal Rule, Simpson's 1/3 Rule, Expression for corresponding error terms	3
4	**Numerical solution of a system of linear equations:** Gauss Elimination Method, Matrix Inversion, LU Factorization Method, Gauss-Seidel Iterative Method	6
5	**Numerical solution of Non-Linear equation:** Bisection Method, Regula- Falsi Method, Newton-Raphson Method	4
6	**Numerical solution of ordinary differential equation:** Euler's Method, Runge-Kutta Methods, Predictor-Corrector Methods, Finite Difference Method	5
7	**Measure of Central Tendency and Dispersion:** Mean, median, modeand S.D.	3
8	**Curve Fitting by Method of Least Square:** Fitting a straight line of the form $y = a + bx$, Fitting a curve of the form $y = ax + bx^2$, $y = ab^x, y = ae^{bx}, y = ax^b$.	3
Total		**30**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Calculate different types of error involved in Engineering problems and learn to apply numerical methods to find approximate solutions for them.
2	Interpolate different polynomials using numerical techniques .
3	Derive numerical methods for integration and apply them for solving otherwise intractable Engineering problems
4	Solve system of linear equations and to learn the concept of root finding for nonlinear equations
5	Solve ordinary differential equation numerically
6	Use various statistical tools to solve Engineering problems numerically

Learning Resources:	
1	C.Xavier: C Language and Numerical Methods.
2	A. K. Jalan and Utpal Sarkar, Numerical Methods-A Programming Based Approach, Orient Blackswan Private Ltd.
3	Dutta & Jana: Introductory Numerical Analysis.
4	J.B.Scarborough: Numerical Mathematical Analysis.
5	Jain, Iyengar , & Jain: Numerical Methods (Problems and Solution).
6	Balagurusamy: Numerical Methods, Scitech.
7	Baburam: Numerical Methods, Pearson Education.
8	N. Dutta: Computer Programming & Numerical Analysis, Universities Press.
9	Soumen Guha & Rajesh Srivastava: Numerical Methods, OUP.
10	Srimanta Pal: Numerical Methods, OUP.

Course Name:	**Software Engineering**		
Course Code:	PE-CS(D)501A	**Category:**	Professional Elective
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	Programming for Problem Solving
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	Exposure of the challenges of large scale software development and the ideas of how to overcome those.
2	To provide the idea of decomposing the given problem into Analysis, Desing, Implementation, Testing and Maintenance phases.
3	To provide an idea of using various process models in the software industry according to given circumstances.
4	To gain the knowledge of how Analysis, Design, Implementation, Testing and Maintenance processes are conducted in a software project.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Module I:** Overview of System, System Development Life Cycle, Waterfall Model , Iterative Waterfall Model, V Model, Spiral Model, Feasibility Analysis, Technical Feasibility, Cost- Benefit Analysis, COCOMO model, SRS.	8
2	**Module II:** System Design – DFD, Top-Down and Bottom-Up design; Decision tree, Decision table and structured English; Functional vs. Object- Oriented approach.	10
3	**Module III:** Coding & Documentation guidelines, Errors, Faults and Failures. Testing – Levels of Testing, Unit Testing, Integration Testing, System Testing, Validation & Verification in Testing, Black Box Testing, White Box Testing Strategies, Equivalence class Testing, Cyclomatic complexity. Introduction to Agile	10
4	**Module IV:** UML diagrams: Class diagram, interaction diagram, collaboration diagram, sequence diagram, state chart diagram, activity diagram.	8
Total		**36L**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Identify a suitable software development life cycle model for an application and prepare software requirements specification and cost estimation for an application.
2	Use the software design concepts like DFD, Top-Down and Bottom-Up design, Decision tree, Decision table to develop a high-quality software.
3	Classify the degree of functionalities and the relationship of modules of a software system.
4	Explain different types of Validation & Verification in testing methods to identify errors during software development and various Object-based view of Systems like Class diagram, interaction diagram, collaboration diagram, sequence diagram, state chart diagram, activity diagram.

Learning Resources:	
1	Rajib Mall, Fundamental of Software Engineering, PHI Learning Pvt. Ltd.
2	Roger S. Pressman, Bruce R. Maxim, Software Engineering : A practitioner's approach– Pressman, McGraw Hill Education
3	Pankaj Jalote , Software Engineering: A Precise Approach, Wiley

Course Name:	**Advanced Algorithm**		
Course Code:	PE-CS501B	**Category:**	Professional Elective
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	Design and Analysis of Algorithm, Data Structure, Discrete Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	Introduce students to the advanced methods of designing and analyzing algorithms.
2	The student should be able to choose appropriate algorithms and use it for a specific problem.
3	To familiarize students with basic paradigms and data structures used to solve advanced algorithmic problems.
4	Students should be able to understand different classes of problems concerning their computation difficulties.
5	To introduce the students to recent developments in the area of algorithmic design.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Sorting:** Review of various sorting algorithms, Topological sorting **Lower Bound Theory:** O(n log n) bound for a comparison sort. **Graph:** Definitions and Elementary Algorithms: Shortest path by BFS, shortest path in edge-weighted case (Dijkasra's), depth-first search and computation of strongly connected components, emphasis on correctness proof of the algorithm and time/space analysis, example of amortized analysis.	5
2	**Matroids:** Introduction to greedy paradigm, algorithm to compute a maximum weight maximal independent set. Application to MST. **Disjoint Set Manipulation:** Set manipulation algorithm like union-find, union by rank, path compression. **Graph Matching:** Algorithm to compute maximum matching. Characterization of maximum matching by augmenting paths, Edmond's Blossom algorithm to compute augmenting path.	6
3	**String matching problem:** Different techniques – Naive algorithm, string matching using finite automata, and Knuth, Morris, Pratt (KMP) algorithm with their complexities **Matrix Computations:** Strassen's algorithm and introduction to divide and conquer paradigm, inverse of a triangular matrix, relation between the time complexities of basic matrix operations, LUP-decomposition.	6

4	**Modulo Representation of integers/polynomials:** Chinese Remainder Theorem, Conversion between base-representation and modulo-representation. Extension to polynomials. Application: Interpolation problem. **Discrete Fourier Transform (DFT):** In the complex field, DFT in modulo ring. Fast Fourier Transform algorithm. Schonhage-Strassen Integer Multiplication algorithm.	5
5	**Computational Geometry:** Robust geometric primitives, Convex Hull, Triangulation, Voronoi diagrams, Nearest neighbor search, Range search, Point location, Intersection detection, Bin Packing, Medial-axis transform, Polygon partitioning, Simplifying Polygons, Shape Similarity, Motion Planning, Maintaining line arrangements, Minkowski sum.	5
6	**Linear Programming:** Geometry of the feasibility region and Simplex algorithm **NP-completeness:** Examples, proof of NP-hardness and NP-completeness.	5
7	Recent trends in problem solving paradigms using recent searching and sorting techniques by applying recently proposed data structures.	4
Total		**36**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Recall lower bound theorem, various graph algorithms along with different sorting algorithms.
2	Explain matroids, disjoint set manipulation, graph matching algorithm, different string matching algorithms and operations on Strassen's matrix manipulation, different DFT algorithms and Modulo representation of Integer/ Polynomial etc.
3	Solve a given problem using appropriate algorithm.
4	Solve given problems using Convex hull, Voronoi diagram, Range search, Bin packing and other methods under computational geometry.

Learning Resources:	
1	"Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein.
2	"The Design and Analysis of Computer Algorithms" by Aho, Hopcroft, Ullman.
3	"Algorithm Design" by Kleinberg and Tardos.
4	Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House, New Delhi.

Course Name:	**Compiler Design**		
Course Code:	PE-CS(AM)501C	**Category:**	Professional Elective
Semester:	Fifth	**Credit:**	3
L-T-P:	3-0-0	**Pre-Requisites:**	Basic programming Knowledge, Automata Theory
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To understand and list the different stages in the process of compilation.
2	Identify different methods of lexical analysis
3	Design top-down and bottom-up parsers
4	Identify synthesized and inherited attributes
5	Develop syntax directed translation schemes
6	Develop algorithms to generate code for a target machine

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Introduction: Programs, interpreters, and translators; Analysis-Synthesis model of translation; Examples of translators; Structure of a compiler; Issues in compiler design.	2L
2	Lexical Analysis: Role of a lexical analyzer; Input buffering, Specification of tokens, Recognition of tokens; Languages, Regular expressions, Regular definitions; Finite automata, Nondeterministic and deterministic finite automata, Transitions tables, Acceptance of input strings by automata, Conversion of an NFA to DFA; State-machine-driven lexical analyzers and their implementations	8L
3	Syntax Analysis: Role of a parser, Representative grammars, Context-free grammars, Parse trees, derivations and sentential forms, Ambiguity; Top down parsing, Predictive and Recursive descent parsing, Elimination of left recursions, Left factoring, FIRST and FOLLOW sets and their computations, LL(1) grammars, Error recovery techniques; Bottom up parsing, Reductions, Handle pruning, Shift reduce parsing; LR parsing, Implementing the parser as a state machine, viable prefixes, Items and the LR(0) automaton; Constructing SLR parsing tables: LR(0) grammars, SLR(1) grammars; Canonical LR(1) items and constructing canonical LR(1) parsing tables; Constructing LALR parsing tables.	10L
4	Semantics and Semantic Analysis: Syntax-directed translation, Attribute grammars, Inherited and synthesized attributes, Dependency graphs, Evaluation orders of attributes, S-Attributed definitions, L-attributed definitions, Syntax-directed translation schemes.	3L
5	Intermediate Code Generation & Runtime Environment: Intermediate languages, Graphical representation, Three-address code, Implementation of three address statements (Quadruples, Triples, Indirect triples). Intermediate languages – Declarations – Assignment Statements – Boolean Expressions – Case Statements – Back patching – Procedure calls. Source language issues (Activation trees, Control stack, scope of declaration, Binding of names), Storage organization	6L

	(Subdivision of run-time memory, Activation records), Storage allocation strategies, Parameter passing (call by value, call by reference, copy restore, call by name), Symbol tables, dynamic storage allocation techniques	
6	Code Optimization: Overview of optimization; Data Flow Analysis; Peephole Optimizations; Constant Folding, Common Subexpression Elimination, Copy Propagation, Strength Reduction. Global Optimization: Loop optimizations; Induction Variable elimination, Optimizing procedure calls – inline and closed procedures. Machine-Dependent Optimization: Pipelining and Scheduling	5L
7	Code Generation: Issues in the design of code generator – The target machine, Construction of executable code and libraries.	2L
Total		**36L**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Summarize the basic concept of compiler and underlying finite state automata, regular expression, grammars and regular languages.
2	Describe the functional phases of a compiler such as lexical analyzer, parser, code optimizer and code generator
3	Compare LL, LR(0), LR(1) and LALR parser.
4	Write semantic rule, quadruple, triples, indirect triple of given expression and optimized code.

Learning Resources:	
1	Aho, Sethi, Ullman - "Compiler Principles, Techniques and Tools" - Pearson Education
2	Holub - "Compiler Design in C" - PHI.
3	"Crafting a compiler with C", C. N. Fischer and R. J. LeBlanc, Pearson Education
4	"Compiler Construction: Principles and Practice", Kenneth C. Louden, , Thomson Learning

Course Name:	**Machine Learning Lab**		
Course Code:	PC-CS(D)591	**Category:**	Professional Core
Semester:	Fifth	**Credit:**	1.5
L-T-P:	0-0-3	**Pre-Requisites:**	BS-M 305
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:	
1	To learn the concepts of data and patterns
2	To design and implement various machine learning algorithms.
3	Implement supervised and unsupervised machine learning
4.	Implement Deep Learning Techniques and various feature extraction with case study.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Implementation of Linear Regression, logistic regression, multivariate logistic regression	9
2	Decision Tree and Ensemble Techniques, Bagging, Boosting	6
3	LSTM algorithm-Time Series Analysis	3
4	Clustering Techniques Implementation – Different Techniques, SVM implementation	6
5	Simple neural network implementation, Deep Learning techniques – Shallow and Deep Neural network Concept of Keras , Tensor Flow	9
6.	Factorization and Generative Model implementation	3
Total		**36**

Course Outcomes:

After completion of the course, students will be able to:

1.	Implement different Regression analysis for single variate and multivariate optimization as well as different classification algorithm.
2.	Implement Decision tree algorithm, Ensemble Techniques, LSTM algorithm.
3.	Implement different clustering and classification algorithm with high end application like SVM.
5.	Implement Deep Learning concepts, Factorization and Generative Models.

Learning Resources:

1	Practical Machine Learning Released January 2016, Publisher(s): Packt Publishing ISBN: 9781784399689 – Sunita Gollapudi
2	Hands-On Machine Learning with Scikit-Learn and TensorFlow- Aureillen Garon O Reilley
3	Hands-On Deep Learning Algorithms with Python-Sudharsan Ravichandiran Packt Publishing
4	Introduction to Machine Learning with Python,by Andreas C. Müller, Sarah Guido Released October 2016, Publisher(s): O'Reilly Media, Inc., ISBN: 9781449369415

Course Name:	**Computer Networks Lab**		
Course Code:	PC-CS(AM)593	**Category:**	Professional Core Courses
Semester:	Fifth	**Credit:**	1.5
L-T-P:	0-0-3	**Pre-Requisites:**	Operating System, Programming knowledge
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 5

Course Objectives:	
1	To develop an understanding of modern network design and configuration methodologies.
2	To introduce the student to the major concepts involved in network communication.
3	To apply the communication concepts in network programming

Course Contents:

Module No.	**Description of Topic**	**Contact Hrs.**
1	Familiarization with network cables (CAT5/5e/6 UTP), connectors (RJ45, T-connector), Ethernet driver, connecting devices (Hubs, Switches, Routers) etc. and preparation of crossover and/or straight-through patch cable using color code.	3
2	System Configuration (Windows and/or Linux) with implementation of subnetting. Familiarization of network related commands like ping, netstat, ifconfig/ ipconfig, netconfig, traceroute, telnet, finger, iptables, ipchains etc.	6
3	Familiarization of network simulator and/or packet tracer, configuration of router and configuration of DNS, FTP, HTTP, Mail server etc.	3
4	Implementation of IPC using Pipe.	3
5	Implementation of IPC using connection-oriented (TCP) and connection-less (UDP) socket in both iterative and concurrent (multi-process and/or multi-threaded approach) modalities.	18
6	Implementation of Data Link Layer Flow Control Mechanism (Sliding Window).	3
Total		**36**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Prepare network cable to connect devices for network design.
2	Apply configuration knowledge and skill to setup Ethernet Card in Windows and Linux.
3	Implement of Inter-Process Communication using Pipe.
4	Design variety of iterative and concurrent servers to implement client-server communication using TCP and UDP socket.

Learning Resources:	
1	"Unix Network Programming: The Sockets Networking API Vol 1" – W. Richard Stevens, Bill Fenner, Andrew M. Rudoff, Third Edition, Addison Wesley
2	"UNIX Network Programming: Interprocess Communications, Volume 2" – W. Richard Stevens, 2nd Edition, Prentice Hall
3	"Hands-On Network Programming with C: Learn socket programming in C and write secure and optimized network code"- Lewis Van Winkle, Packt Publishing Limited

Course Name:	**Artificial Intelligence Lab**		
Course Code:	PC-CS(AM) 592	**Category:**	Professional Core Courses
Semester:	Fifth	**Credit:**	1.5
L-T-P:	0-0-3	**Pre-Requisites:**	Data Structure & Algorithm, Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:	
1	To implement different Searching and Gaming Strategies
2	To implement Probabilistic Reasoning Techniques
3	To implement different Logical Programming
4.	To implement Natural Language Processing

Course Contents:

Module No.	**Description of Topic**	**Contact Hrs.**
1	Implementation of some problems related with Production System, BFS, DFS with Python/R	3
2	Implementation of Heuristic Search Strategies using Python/R (Any 2 search strategies)	3
3	Games Designing with AI Techniques using Python/R (Using MINIMAX)	3
4	Probabilistic Reasoning with Sequential Data	3
5.	Logic Programming using PROLOG/LISP and difference in Python/R implementation	15
6.	Introduction to NLTK- Handling Natural Language Processing	9
Total		**36**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Implement different searching and gaming strategies using AI Techniques.
2	Implement Probabilistic Reasoning Techniques with sequential data.
3	Implement Logic Programming with Prolog /LISP and modern Programming Languages like Python/R.
4.	Implement Natural Language Processing Techniques using Tool kit.

Learning Resources:	
1	Artificial Intelligence with Python – Prateek Joshi PACKT Publishing
2	LISP- Patrick Henry Winston, Berthhold Klaus Paul Horn, Published by Pearson
3.	Logic and Prolog Programming by Saroj Kaushik Published by New Age International
4.	Prolog Programming for Artificial Intelligence by Ivan Bratko, Published by Pearson

Course Name:	**Aptitude Skill Development-I**		
Course Code:	MC571	**Category:**	Mandatory Courses
Semester:	Fifth	**Credit:**	0
L-T-P:	2-0-0	**Pre-Requisites:**	Basic knowledge of Mathematics and English Language
Full Marks:	100		
Examination Scheme:	Semester Examination: 100	Continuous Assessment: 00	Attendance: 00

Course Objectives:	
1	To be familiar with the basic concepts of QUANTITATIVE ABILITY.
2	To be familiar with the basic concepts of LOGICAL REASONING Skills .
3	To be familiar with the basic concepts of PROBABILITY.
4	Acquire knowledge in VERBAL REASONING and VOCABULARY

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Basics of Quantitative Abilities: Number System, HCF and LCM, Average, Ratio, Proportion and Variations, Problems on Percentage.	**4L**
2	Arithmetic Quantitative Abilities: Problems on Ages, Profit and Loss, Time and Work, Problems on Simple and Compound Interest, Problems on Time, Speed and Distance.	**6L**
3	Permutation and Combination, Set theory, Mensuration and Logarithm.	**5L**
4	Logical Reasoning: Number Series, Alpha Numerical, Letter & Symbol Series, Numerical and Alphabet Puzzles, Seating Arrangement, Blood Relation and Calendars.	**7L**
5	Data Interpretation	**2L**
Total		**24L**

Course Outcomes:

After completion of the course, students will be able to:

1	Understand the basic concepts of QUANTITATIVE ABILITY.
2	Understand the basic concepts of LOGICAL REASONING Skills .
3	Understand the basic concepts of PROBABILITY.
4	Acquire satisfactory competency in use of VERBAL REASONING

Learning Resources:

1	Arun Sharma, “Quantitative abilities”, McGraw-Hill
2	R.S.Agrawal, “Quantitative Aptitude for Competitive Examinations”, S. Chand
3	R.S.Agarwal,”A Modern Approach to Verbal & Non-Verbal Reasoning ”,S.Chand